INCH-POUND

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DETAIL SPECIFICATION

ARMOR PLATE, STEEL, METALLURGICALLY BONDED, DUAL HARDNESS (0.187 INCHES TO 0.700 INCHES INCLUSIVE)

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers metallurgically bonded, dual hardness, steel armor plate for ordered thicknesses from 0.180 inch to 0.700 inch for lightweight armor applications. (see 6.1, 6.2, and 6.3.17).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

USADTC TOP 2-2-710	-	Ballistic Tests of Armor Materials
ITOP 2-2-713	-	Ballistic Testing of Armor

(Copies of this document are available online at https://discover.dtic.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF NONDESTRUCTIVE TESTING

ANSI/ASNT CP-189-2016	-	ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel

(Copies of this document are available online at https://www.asnt.org/store.)

ASTM INTERNATIONAL

ASTM A6/A6M	-	Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes and Sheet Piling
ASTM A370	-	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A578/ A578M	-	Standard Specification for Straight-Beam Ultrasonic Examination of Plain and Clad Steel Plates for Special Applications
ASTM A751	-	Standard Test Methods, Practices, and Terminology for Chemical Analysis of Steel Products
ASTM E10	-	Standard Test Method for Brinell Hardness of Metallic Materials

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ASTM E18	-	Standard Test Methods for Rockwell Hardness of Metallic Materials
ASTM E110	-	Standard Test Method for Indentation Hardness of Metallic Materials by Portable Hardness Testers
ASTM E350	-	Standard Test Methods for Chemical Analysis of Carbon Steel, Low-Alloy Steel, Silicon Electrical Steel, Ingot Iron, and Wrought Iron
ASTM E1077	-	Standard Test Methods for Estimating the Depth of Decarburization of Steel Specimens
ASTM E1444	-	Standard Practice for Magnetic Particle Testing

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2, 6.5 and 6.9), a sample or samples of the specified item shall be made available to the contracting officer or their authorized representative for approval in accordance with 4.4. The contractor shall comply with this requirement at the time of their first order or contract and at any time that the supplier has not furnished the dual hardness armor plate in the applicable thickness range under this specification within a period of 37 months. The approval of the first article samples authorizes the commencement of shipment but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification, namely conformance or production acceptance. The first article samples and test plates shall be manufactured by the process proposed for use on production armor. This process shall be inspected by a cognizant government representative to ensure compliance with the details of this specification. The manufacturer's declared chemical analysis shall be submitted to the contracting agency and to the ballistic test agency. The ballistic test agency shall record the first article ballistic test plates submitted, showing the dates tested. Requests from the procuring activity to the ballistic test agency as to prior conformance with first article tests shall be accompanied by copies of the first article test firing records. Any deviation(s) noticed by the ballistic agency shall be brought to the attention of the contracting activity and to the manufacturer. After first article testing has been accepted by the contracting activity, the composition and processing shall be considered fixed. If either of these parameters change outside of the limitations allowed, then a new first article test shall be required.

3.1.1 First time producer. First time producers wishing to qualify to this specification shall follow the instructions of 6.6.

3.2 Production acceptance.

3.2.1 Chemical composition. Unless otherwise specified (see 6.2d), the chemical analysis of the dual hardness armor plate shall be within the composition limits established by the contractor. A declared chemistry shall be submitted to the contracting agency or its authorized representative

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and to the ballistic test agency. If an analysis of antimony, arsenic, or lead is required, it shall be specified in the contract or purchase order (see 6.2e). The chemical composition shall be determined by a product analysis (values shall be listed as weight percent) in accordance with 4.4.1.2 and 4.5.1.

3.2.2 Carbon equivalence. Carbon equivalence (CE) shall be calculated and recorded (see 6.2f), for each layer of each heat per ASTM A6/A6M, i.e.

$$CE = C + [Mn/6] + [(Cr + Mo + V)/5] + [(Ni + Cu)/15]$$

where the elements are expressed in wt%.

The CE values for each layer shall be reported on the completed Dual Hardness Armor (DHA) Test Data Form (see figure 1).

3.3 Material and processing. The chemical composition of the steel, the heat treatment, and methods of fabrication will be left entirely to the decision of the armor manufacturer. The process of manufacture and the heat treatment shall be such as to produce metallurgically bonded armor that will meet the requirements of this specification. No operation that may alter the ballistic characteristics of the armor shall be employed after the ballistic test plate representing a lot has been selected.

3.4 Hardness. The hardness range of the armor plate surfaces (table I) when tested as specified in 4.5.2 and 4.7.2.2 shall be as follows:

TABLE I. Hardness requirements.

Armor Face	Hardness Range (min - max)	
	Rockwell C	Brinell Hardness
Front	58 - 64	601 - 712
Rear	48 - 54	461 - 534

Unless otherwise specified the cross-section hardness shall be determined and shall comply with the purchaser's requirements (see 4.5.2 and 6.2j).

3.5 Decarburization. The manufacturer shall produce with a practice designed to minimize complete and partial decarburization (6.2k).

3.6 Bonding. The armor plate shall be composed of metallurgically bonded plates. Ultrasonic evaluation shall be conducted to locate debonded areas (see 4.7.2.4). Metallographic evaluation shall be conducted on corner samples to evaluate bond effectiveness and on samples of debonded areas located by ultrasound. The presence of bond defects shall be reported to the purchaser. When specified, (see 6.2l) additional testing or examination of bonds shall be conducted as agreed upon by the contractor and purchaser. Rejection of armor plate because of bond defects shall be as agreed upon by the purchaser and supplier (see 6.2l).

3.6.1 Internal soundness and soundness of bonding. Plates shall be ultrasonically tested unless supplier can prove substantial plates have been ultrasonically tested in the past without findings.

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If supplier can demonstrate excellent soundness history, then only one plate from every lot shall be ultrasonically examined for internal soundness and soundness of bonding in accordance with 4.7.2.4 The acceptance level shall be C as defined in ASTM A578/A578M, unless otherwise specified in the contract or purchase order (see 6.2m).

3.7 Ballistic requirements. The ballistic requirements of the armor plate shall be in accordance with Appendix A of this specification. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be utilized. When four (4) partial penetrations are above the minimum requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be specified) above the minimum requirement.

3.7.1 Ballistic test plate information. For each lot of armor plate a properly completed Dual Hardness Armor (DHA) Test Data Form (see figure 1) shall be submitted to the ballistic testing facility (see 4.2.1 and 6.2l) with each ballistic test plate that represents that particular processing lot. If another form, such as a Material Test Report or a Test Certificate, that contains the same information as in figure 1, is available, it can be used in combination with figure 1, unless otherwise specified in the contract or purchase order (see 6.2). This additional form shall be attached to figure 1 and submitted with each ballistic test plate.

3.8 Finish. Unless otherwise specified in the contract or purchase order (see 6.2n), plates shall be descaled after treatment. Plates shall not be pickled.

3.9 Edge preparation. Thermal cutting shall be permitted after final heat treatment provided the procedure, which may include grinding after thermal cutting, is such that no cracks develop on any thermally cut edge whether detected by nondestructive inspection, or as agreed upon in the contract or purchase order.

3.10 Dimensions and tolerances. The dimensions shall be as specified in the contract or purchase order (see 6.2c, 6.2g - 6.2i). Unless otherwise specified or permitted in the contract or purchase order, the following tolerance shall apply:

Flatness:	Within 0.75 inch from a straight line 6 feet in length.
Thickness:	+/- 0.016 inch for plate thicknesses 0.187 to 0.500 inch inclusive and +/- 0.019 inch for plate thicknesses over 0.500 through 0.700 inch.

3.11 Identification Marking. Identification marking and Dual Hardness Armor (DHA) Test Data Form (see figure 1) records shall be such as to ensure positive identification of all plates, including test samples and specimens, with the lot and corresponding heat from which they were produced. The primary plate rolling direction shall be identified. Each plate shall be marked with the DLA contract number, the NSN (if applicable), manufacturer's name or trademark, the number of this specification, the plate thickness in inches and the lot number. The marking shall be such as to identify the higher hardness side. Unless otherwise specified (see 6.2o) or permitted on the contract or purchase order, impression stamping shall not be used except for proper heat and hard-soft side identification during the rolling process. If impression stamping is permitted, it shall be applied on the lower hardness surface only. First article and acceptance ballistic test plates shall also be marked with the manufacturer's name or trademark, the number of this specification, and the ordered plate thickness in inches. First article plates shall be marked "PRE," acceptance plates

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“ACC," and retest plates shall be marked “Rl” and “R2.” If a second set of retest plates are submitted they shall be marked “RR1” and “RR2.”

3.12 Workmanship.

3.12.1 Surface imperfections. When specified in the contract or purchase order (see 6.2), plates shall be inspected for surface imperfections in the de-scaled condition, only. The front and rear surface of each plate shall be free from the following surface defects: slivers, laps, checks, seams, blisters, snakes, cold shuts, cracks, burning, and other defects of such severity that are detrimental to fabrication or performance of the armor plate (see 6.3.6 - 6.3.17). Imperfections listed above which are of such nature as to affect the fabrication of the material, are rejectable.

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REQUEST FOR BALLISTIC TEST OF DUAL HARDNESS ARMOR																	
SPECIFICATION: MIL-DTL-46099						**REVISION:**					**AMENDMENT:**						
FIRING RECORD:						**DATE:**					**TEST CONTRACT NO:**						
	PLATE/COIL MELTED					**PLATE/COIL ROLLED**					**PLATE/COIL HEAT TREATED**						
Name:																	
Address:																	
POC:																	
Phone No:																	
Fax No:																	
TEST ITEM IDENTIFICATION:											**Ordered Thickness:** _____						
Lot No._____ **Plate No.** _____					**NSN:** _____ (as applicable)						**SURFACE & WORKMANSHIP:** **(Pass / Fail)** _____						
PURPOSE: ___ Acceptance ___ First Article ___ Reduced Testing																	
SAMPLE: ____ Primary ____ Retest (Firing Record No. of Failed Sample_____)																	
CHEMISTRY ANALYSIS	**Side**	C	Mn	P	S	Si	Ni	Cr	Mo	V	Nb	B	N	Cu	Ti	Al	
FIRST ARTICLE:	Hard																
	Soft																
PRODUCTION:	Hard																
	Soft																
ACCEPTANCE: (Pass / Fail)	Hard																
	Soft																
CARBON EQUIVALENCE	**HARD LAYER (CE)** ____								**SOFT LAYER (CE)** ____								
HARDNESS READINGS:	**Hard Face: 1:**_____ **2:**_____ **3:**_____								**Soft Face: 1:**_____ **2:** _____ **3:**_____								
BALLISTIC TEST RESULTS: SEPARATELY HEAT TREATED TEST PLATE : YES_____**NO**_____																	
	Obl. (deg)	**Actual Thickness (in)**			**Required V_{50} (fps)**				**Actual V_{50} (fps)**						**Pass/ Fail**		
Test:																	
Projectile:																	
	NOTES:																
Lot [met] [failed to meet] the ballistic requirements of specification MIL-DTL-46099																	
Government Representative:				**Date:**		**Supplier Representative:**								**Date:**			

FIGURE 1. Dual Hardness Armor (DHA) Test Data Form.

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4. VERIFICATION

4.1 Testing responsibility and facilities. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all the requirements as specified herein. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor may use their own or any other facilities suitable for the performance of the requirements specified herein, except ballistic tests (see 4.2.1), unless disapproved by the Government. The Government reserves the right to perform or check any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements and to determine the validity of the certifications.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract or purchase order. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4.1).
b. Production acceptance inspection (see 4.5).

4.2.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2z), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

4.3 Lot. A lot shall consist of all production and ballistic test plates of the same melt of steel, of the same thickness, having the same treatment, manufactured and heat treated with the same thermal cycle in the same production furnace(s) in the same facility under the same time periods. When specified by the procuring activity (see 4.3.1 and 6.3.5), production and ballistic test plates shall be allowed to be heat treated separately. The test plate shall be heat treated in a production furnace.

4.3.1 Separately heat treated ballistic test plate. When the procuring activity allows a ballistic test plate to be heat treated separately from the production plates it represents (see 4.3), it shall be so stated in the data (see 6.6z).

4.4. Sampling.

4.4.1 First article sampling.

4.4.1.1 Ballistic test samples. Two ballistic test plates for each nominal thickness range (see table II) to be supplied on the contract or purchase order, shall be submitted for ballistic tests. One sample shall be taken from the first plate fabricated and one from the last plate fabricated in the

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initial lot produced. When only one plate is fabricated, the samples shall be taken from opposite ends of the plate. The samples shall be 12 inches by 36 inches and shall be forwarded to the Government facility selected by the procuring activity (see 6.3.5).

TABLE II. Ballistic test plate sizes and corresponding test projectiles for first article and acceptance testing.

Nominal thickness range, inches	Minimum size 1/ of test plates, inches	Test projectile and obliquity	TABLE
0.180 to 0.3125 incl.	12 x 36	caliber .30 M2 AP at 30°	A-I
0.3126 to 0.565 incl.	12 x 36	caliber .50 M2 AP at 30°	A-II
Over 0.565 incl.	TBD	TBD	TBD

1/ Other sizes may be submitted for the convenience of the manufacturer if approved by the Survivability/Lethality Directorate, USATC, Aberdeen Proving Ground.

4.4.1.2 Samples for chemical composition. Two samples for chemical analysis shall be taken from near each plate selected for the ballistic tests in accordance with ASTM E350. Samples shall be taken from the hard and soft sides of the armor plate, and each shall be identified accordingly.

4.5 Production acceptance inspection. The tests under 4.7 and the retest procedures under 4.8 shall serve as a basis for the acceptance of individual production lots.

4.5.1 Chemical composition. Representative samples shall be taken for chemical analysis from each lot in accordance with ASTM E350. Samples shall be taken from the hard and soft sides of the armor plate, and each shall be clearly identified.

4.5.2 Surface hardness. The location and number of samples to be taken from each lot shall be in accordance with applicable drawings or as otherwise specified (see 6.2j).

4.5.3 For microscopic examination. Two samples, each of the full thickness of the annealed plate and with both front and rear surfaces preserved shall be taken from each lot. Samples shall be approximately 1 X 3 inches and shall be taken from diagonal corners of an armor plate representative of the rolling operation. One sample only will be required from one end when a single plate is rolled. (6.2k - 6.2m)

4.5.4 Sampling for ballistic testing. One plate shall be taken from each lot. The plate size shall be 12 inches by 36 inches.

4.6 Examination.

4.6.1 Visual. All steel plate shall be subject to visual inspection for compliance with the requirements for surface conditions (see 3.12.1), edge quality (see 3.9) and identification marking (see 3.11).

4.6.2 Dimensions. All steel plates shall be subject to inspection for compliance with dimensional and tolerance requirements (see 3.10, 6.2c, and 6.2g - 6.2i).

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4.6.3 Packaging. Prior to shipment, examination shall be made to determine compliance with the requirements of section 5.

4.7 Tests.

4.7.1 First article and quality conformance ballistic testing. Ballistic test plates shall be tested at a Government facility unless otherwise specified in the contract or purchase order (see 6.2z). Plate thicknesses as determined by the ballistic test agency shall be used to determine the required protection ballistic limit for the plate and shall be taken as the average of 4 thickness measurements read to the nearest 0.001 inch with the average reported to the nearest 0.001 inch. At least one measurement will be taken along each edge of the plate at a distance of at least one inch from the edge. Testing shall be conducted in accordance with the requirements of the Appendix A.

4.7.1.1 Surface hardness test on ballistic plates. Surface hardness of ballistic test plates shall be determined in accordance with 4.7.2.2.

4.7.2 Quality conformance tests.

4.7.2.1 Chemical analysis. Chemical analysis shall be conducted in accordance with ASTM E350, or other approved method analytical method. In case of dispute, the analysis as presented in ASTM E350 shall be the basis for acceptance or rejection.

4.7.2.2 Rockwell hardness or Brinell hardness tests. Hardness test specimens shall be prepared and tested in accordance with ASTM A370. At least three hardness tests shall be taken on each surface and the average value determined for each surface of each ballistic plate.

4.7.2.3 Microscopic examination. Specimens for microscopic examination shall be suitably mounted and etched to reveal the microstructure of the cross section perpendicular to the rolling direction. Examination shall be conducted at X100. The etched surface of the annealed armor plate shall be examined for decarburization and bonding defects (see 3.5 and 3.6).

4.7.2.4 Ultrasonic examination.

4.7.2.4.1 Inspection equipment. The ultrasonic soundness inspection equipment shall conform to ASTM A578/A578M.

4.7.2.4.2 Procedure. Unless otherwise specified (see 6.2t), the ultrasonic examination (see 3.6) shall be carried out in accordance with ASTM A578/A578M (see 3.2.11.4) with the following exceptions:

a. Scanning shall be continuous over 100% of the plate surface with not less than 10% overlap between each path.
b. Scanning rate shall be at a speed where recordable discontinuities can be detected.
c. The testing frequency shall be a minimum of 2 megahertz (MHz).
d. Any area within a plate where a discontinuity produces a continuous total loss of back reflection accompanied by continuous indications on the same plane +/- 5% of plate thickness? that cannot be encompassed within a circle whose diameter is 1 inch shall be cause for rejection of that plate.

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4.7.2.4.3 Certification of inspection personnel. Unless otherwise specified in the contract or purchase order (see 6.2u), personnel performing ultrasonic inspection shall comply with the qualification requirements of ANSI/ASNT CP-189-2016.

4.8 Rejection and retests.

4.8.1 Rejection. Unless otherwise specified (see 6.2x) where one or more test specimens fail to meet the requirements of the specification the lot represented by the specimen or specimens shall be subject to the rejection.

4.8.2 Retest. When no sampling plan is provided or approved by the procuring agency (see 6.2v) and where there is evidence that indicates that the specimen was not representative of the lot of material, and when the detail specification does not otherwise specify, at least two specimens shall be selected to replace each test specimen that failed. All specimens so selected for retest shall meet the requirements of the specification or the lot shall be subject to rejection.

4.8.3 First article ballistic test failure. In the event of a first article ballistic test failure the supplier must submit two new first article test samples as in 4.4.1.1 and both must pass the ballistic test requirements in order that the first article armor plate be considered ballistically acceptable.

4.9 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced (see 6.2w) provided the results on consecutive lots indicate that a satisfactory uniform product meeting the requirements is being produced. Reduced testing shall be in accordance with a system previously approved or established by the procuring activity involved.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2y). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The metallurgically bonded steel armor specified herein is intended for use in aircraft and lightweight vehicular applications where maximum resistance to armor piercing types of ammunition and multiple hit capability are required. It is normally furnished in thicknesses of 0.187 inch to 0.700 inch. Although the primary intent of DHA is armor appliqué, DHA could be utilized as a welded structural plate if welded by special techniques. Because of the high carbon

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equivalence (CE) of the type of steel specified for the harder layer, it will be more difficult to weld than MIL-DTL-46100E.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When first article sample is required but ballistic test agency should not report test results to contractor. (see 3.1, 4.4.1, and 4.4.2).
c. Specify ordered thickness (see 1.1).
d. If the declared chemistry composition can be different (see 3.2).
e. If an analysis of Antimony, Arsenic or Lead is required (see 3.2.1)
f. Calculate and report CE values for each plate layer (see 3.2.2).
g. Specify thickness and dimensional tolerances (see 3.10).
h. If the flatness tolerance is different than those specified (see 3.10).
i. If tighter flatness tolerance requirements are to be specified (see 3.10).
j. Sampling for surface hardness tests (see 3.4 and 4.5.2).
k. Limits on decarburization (see 3.5).
l. Evaluation of bonding requirements and plate acceptance (see 3.6, 3.6.1, and 4.7.2.3).
m. If requested in the contract or purchase order, specify the acceptance level for bonding integrity (see 3.6.1) as agreed upon by the contractor and purchaser.
n. When plates are not to be descaled (see 3.8).
o. When impression stamping is permitted (see 3.11)
p. If inspection for surface imperfections is required (see 3.12.1).
q. Designation of first article test plates (see 4.4.1.1 and 4.7.1)
r. If someone other than the contractor is responsible for the performance of all the requirements of the specification (see 4.1).
s. If the contractor can't use their own facility or any other facility for testing (see 4.1).
t. If ultrasonic examination is to be different (see 4.7.2.4).
u. If certification of inspection personnel is different (see 4.7.2.4.3).
v. Retest requirements if different than specified in 4.8.2.
w. Specify reduced testing plan when applicable (see 4.9).
x. If rejection requirements differ (see 4.8.1).
y. Packaging requirements if different than specified in Section 5.1.
z. If the ballistic tests are to be conducted at another location (see 4.2.1).
aa. If production and ballistic test plates can be heat treated separately (see 4.3.1).
ab. If definition of a crack is different (see 3.12.1 and 6.3.18)

6.3 Definitions.

6.3.1 Contractor. The contractor or prime contractor is the company which has a direct contract from the Government to furnish an end item, usually a vehicle.

6.3.2 Contracting officer. The term "contracting officer' means the person executing a contract on behalf of the Government and any other officer or civilian employee who is properly designated contracting officer; and the term includes, except as otherwise provided, the authorized representative of a contracting officer acting within the limits of his or her authority.

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6.3.3 Dual hardness armor. Dual hardness armor is armor manufactured through metallurgical bonding processing of differing hardness steel precursor plates to produce integral metallurgically bonded armor plate.

6.3.4 Manufacturer. The manufacturer is defined as the company producing the steel alloy plate.

6.3.5 Procuring activity. The term "procuring activity" is that activity of the Government which actually initiates the request for procurement and maintains the records of the procurement.

6.3.6 Slivers. An imperfection consisting of a very thin elongated piece of metal attached by only one end to the parent metal into whose surface it has been worked.

6.3.7 Laps. A surface imperfection with appearance of a seam caused by hot metal, fins or sharp corners being folded over and thus being forged or rolled into the surface but without being welded.

6.3.8 Checks. Checks are numerous very fine cracks at the surface of a metal part. Checks may appear during processing or during service and are most often associated with thermal cycling or thermal treatment. They are also called check marks, checking, and heat checks.

6.3.9 Seams. A seam is an un-welded fold or lap that appears as a crack, usually resulting from a discontinuity on a metal surface.

6.3.10 Blisters. A raised area, often dome shaped, resulting from delamination under pressure of expanding gas trapped in a metal in a near sub-surface zone. Very small blisters may be called pinhead blisters or pepper blisters.

6.3.11 Snakes. Any crooked surface imperfection in a metal plate, resembling a snake.

6.3.12 Cold shuts. Freezing of the top surface of an ingot before mold is full.

6.3.13 Burning. Burning is permanently damaged metal due to overheating enough to cause incipient melting or intergranular oxidations. Note: this condition is usually obscured by normal cleaning methods and would require deep pickling and/or metallography to note the continuous oxidation (chicken wire effect) of the enlarged grain boundaries.

6.3.14 Lamination. A type of discontinuity with separation or weakness generally aligned parallel to direction of the worked surface of the metal and may be the result of pipe, blisters, seams, inclusions, or segregation; elongated and made directional by working.

6.3.15 Pit. A pit is a cavity or depressed area on the surface of a plate.

6.3.16 Linear indication. For nondestructive examination purposes, a linear indication is evidence of a discontinuity that requires interpretation to determine its significance.

6.3.17 Crack. A crack is a planar discontinuity in the metal that has length and is at least 0.002 inch deep. Cracks may be surface cracks, layer cracks, or through cracks. For the purposes of this specification, a crack is identifiable by the unaided eye unless otherwise specified in the contract or purchase order (see 6.2).

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6.3.18 Thickness. The symbol "T" is used throughout this specification to indicate the nominal thickness of the plate under consideration.

6.4 Chemical analysis. Suggested ASTM instrumental methods that can be used for chemical analysis are E322 and E415. ASTM A751 should be consulted for a complete list of methods.

6.4.1 Changes in composition. Changes in composition may not necessarily be cause for retesting but should be subject to review by the procuring activity to determine whether additional first article testing is required.

6.4.2 Carbon equivalence. The CE requirement is for documentation purposes only and there are no set limit requirements for CE for DHA layers. The CE values for each layer should be calculated per ASTM A6/A6M and recorded using the Dual Hardness Armor (DHA) Test Data Form (See figure 1).

6.4.3 Changes from prior chemical analysis requirements. The compositions of elements antimony, arsenic, lead, or zirconium are not required in the chemical composition report.

6.5 Special first article ballistic test. Special first article ballistic tests are required when the manufacturer changes either the melting procedure or heat treatment or the declared chemistry of the armor. Changes to metallurgical bonding procedures will also require special first article ballistic tests.

6.6 Potential suppliers. Potential suppliers who have not previously supplied armor plate to MIL-DTL-46099 and wish to have their material ballistic tested may do so at their own expense. It is recommended that inquiries for such testing be directed to Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059. The armor manufacturing process should be inspected by a cognizant government representative to ensure compliance with the details of this specification.

6.7 New contracts sponsored by Government agencies. At the time that a new contract or purchase order is initiated for the production of combat vehicles or other systems, the contractor's supplier is to estimate the number, size and delivery schedule of the ballistic test plates which are to be submitted for first article or acceptance testing. A lead time of 60 days after the contract has been signed is to be allowed prior to shipment of the first ballistic test plate(s) to APG to ensure that all administrative functions for the establishment of a new ATC project have been completed in preparation for the test. The contracting government agency is to initiate the new project through a letter to Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 requesting a cost estimate for the ballistic testing of the applicable number of sizes of plates. In the case of increases in scope of existing projects, similar correspondence is needed.

6.8 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds and feet per second may be converted to the metric equivalent by multiplying them by the following conversion factors listed in table III.

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TABLE III. Conversion factors. [1/]

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/sec	0.3048	=	meter per second (m/s)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Centigrade (°C)

[1/] Conversion factors can be associated with ASTM E380 entitled “Metric Practice Guide”.

6.9 Subject term (key word) listing.

Ballistic limit
Ballistic plate
Caliber .30 AP M2
Caliber .50 AP M2
Diffusion bonded
Dual Hardness
Explosively bonded
Plate
Roll bonded

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

BALLISTIC TESTING OF ARMOR, STEEL PLATE, METALLURGICALLY-BONDED, DUAL HARDNESS

A.1 SCOPE

A.1.1 Scope. This appendix covers the requirements for ballistic testing of metallurgically bonded, dual-hardness steel armor plate.

A.2 DEFINITIONS

A.2.1 Fair impact. A fair impact results when an unyawed projectile strikes an unsupported area of the ballistic test plate sample at such a location that there is at least three inches of undisturbed metal between this impact and any previous impact, hole, crack, plate edge, or spalled area.

A.2.2 Witness plate. A witness plate is normally a 0.014 inch thick sheet of 5052 aluminum alloy (or a 0.020 inch thick sheet of 2024-T3 aluminum alloy placed 6 inches (+ 1/2 inch) behind and parallel to the test plates or other ballistic sample.

A.2.3 Complete penetration, protection, CP (P). A protection complete penetration occurs when the projectile or one or more fragments of a projectile or plate pass beyond the back of the test plate and perforates the witness plate.

A.2.4 Partial penetration, protection, PP (PI). A partial penetration is any impact that is not a complete penetration.

A.2.5 Gap. A gap is the difference in velocity between the high partial penetration velocity and the low complete penetration velocity used in computing the ballistic limit where the high partial penetration velocity is lower than the low complete penetration velocity.

A.2.6 Protection ballistic limit, BL (P). The protection V_{50} ballistic limit is defined as the average of 6 fair impact velocities comprising the three lowest velocities resulting in complete penetration and the three highest velocities resulting in partial penetration. A maximum spread of 150 feet per second shall be permitted between the lowest and highest velocities employed in determination of ballistic limits. In cases where the lowest velocity resulting in a complete penetration is 150 fact per second or more below the highest velocity resulting in a partial penetration, the ballistic limit will be based on 10 velocities comprising the five lowest velocities which resulted in complete penetration, and the five highest velocities that resulted in partial penetration. When the 10-round, excessive spread, ballistic limit is used, the velocity spread will be reduced to the lowest practical level (as close to 150 fps as possible). When a 10-round ballistic limit is used, this will be noted in all reports. All velocities shall be corrected to striking velocity. In the event that the ballistic limit computed is less than 30 fps above the minimum required and if a gap (high partial penetration velocity below the low complete penetration velocity) of 30 fps or more exists, firing shall continue as needed to reduce this gap to 25 fps or less. (This procedure shall insure better evaluation of the steel when the ballistic limit is near the minimum required.)

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A.2.7 Backspall. Fragment(s) of the back of the plate detached from the plate by projectile impact on the front.

A.2.8 Exit diameter. The average of the minimum and maximum distance across the exit hole made by a complete penetration impact (see figure A-1).

A.2.9 Overall average exit diameter. The overall average exit diameter is defined as the average of the three exit diameters on the three complete penetrations used in the computation of the ballistic limit (see figure A-2).

A.3 REQUIREMENTS

A.3.1 Resistance to penetration. The minimum acceptable ballistic limits shall be in accordance with the values shown in tables A-I through A-II.

A.3.2 Resistance to spalling.

A.3.2.1 Plates 0.187 through 0.400 inch in thickness. The overall average exit diameter, (paragraph A.2.9 and figure A-2), shall not exceed 0.500 inch. The exit holes on the three complete penetrations used in the computation of the six round ballistic limit will be measured.

A.3.2.2 Plates 0.401 through 0.700 inch in thickness. The overall average exit diameter, (paragraph A.2.9 and figure A-2), shall not exceed 0.750 inch. The exit holes on the three complete penetrations used in the computation of the six round ballistic limit will be measured.

A.3.2.3 Measurement of borderline overall average exit diameter. In situations where the overall average exit diameter is borderline, a planimeter will be used to determine the overall average exit diameter (figure A-3).

A.3.3 Rejection. Failure to meet any of the above requirements shall be cause for plate rejection.

A.4 TESTS

A.4.1 Ballistic tests. V_{50} ballistic tests shall be performed in accordance with USATECOM TOP 2-2-710, Ballistic Tests of Armor Materials to determine compliance with the requirements of tables A-I and A-II.

A.4.1.1 Plate thickness. Plate thickness as measured by the ballistic test agency shall be used to determine the required ballistic limit of the plate. The required ballistic limit shall be determined by interpolation of the tables in the appendix, if necessary.

A.4.1.2 Markings. Identifying markings shall be provided on all test and production plates submitted under this specification as specified in 3.9.

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APPENDIX A

A.4.1.3 Rejection and retest of ballistic plates.

A.4.1.3.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of either of the first article test plates to meet the minimum ballistic requirements as specified in the appendix indicates failure of the product and process.

A.4.1.3.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates must be submitted for first article testing and both must pass.

A.4.1.3.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot, however; the final decision will depend on the outcome of retests, if submitted.

A.4.1.3.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer has the following options: Immediately upon notification of the failure, the manufacturer may:

(1) Submit two additional test plates from the same lot at their own expense for ballistic retest, or

(2) The manufacturer may re-heat treat (quenching and tempering) the lot and then submit a test plate from the re-treated lot, or

(3) The manufacturer may scrap the entire lot and submit a plate representing a new lot for acceptance.

If the manufacturer selects any one of these options and the ballistic retest plate (or plates) meet the requirements then the lot represented is acceptable. If the manufacturer selects option (1) and one or both of the retest plates fail, the manufacturer may re-heat treat that lot and submit a test plate from the retreated lot. If this plate fails that lot is rejected. If the manufacturer selects option (3) and the test plate fails, the manufacturer may again resort to any one of the three options.

A.4.1.4 Disposition of ballistic test plates.

A.4.1.4.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates will be returned “as is” to the applicant, at their expense, unless the plates were destroyed in testing.

A.4.1.4.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered as part of the lot of steel they represent and ownership of them passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as the rejectable lot does. The failed plates will be returned, upon request, as in A.4.1.4.1.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits (protection criteria) - caliber .30 AP M2 projectile at 30° obliquity.

Thickness, inches	BL(P), ft/sec	Thickness, inches	BL(P) ft/sec
0.170	2133	0.255	2659
0.175	2167	0.260	2687
0.180 1/	2201	0.265	2715
0.185	2234	0.270	2742
0.190	2267	0.275	2769
0.195	2299	0.280	2796
0.200	2331	0.285	2823
0.205	2363	0.290	2849
0.210	2394	0.295	2875
0.215	2425	0.300	2902
0.220	2455	0.305	2927
0.225	2486	0.310	2952
0.230	2515	0.3125 2/	2965
0.235	2545	0.315	2978
0.240	2574	0.320	3003
0.245	2603	0.325	3028
0.250	2631	0.330	3052

1/ Specification requirements begin for this ordered thickness

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits (protection criteria) - caliber .50 AP M2 projectile at 30° obliquity.

Thickness, inches	BL(P), ft/sec	Thickness, inches	BL(P) ft/sec
0.290	1774	0.440	2565
0.295	1805	0.445	2587
0.300	1836	0.450	2610
0.305	1867	0.455	2632
0.310	1897	0.460	2653
0.3126 1/	1912	0.465	2675
0.315	1927	0.470	2697
0.320	1956	0.475	2718
0.325	1985	0.480	2739
0.330	2013	0.485	2760
0.335	2041	0.490	2781
0.340	2069	0.495	2802
0.345	2097	0.500	2823
0.350	2124	0.505	2843
0.355	2150	0.510	2864
0.360	2177	0.515	2884
0.365	2203	0.520	2904
0.370	2229	0.525	2924
0.375	2254	0.530	2944
0.380	2280	0.535	2963
0.385	2305	0.540	2983
0.390	2330	0.545	3002
0.395	2354	0.550	3022
0.400	2378	0.555	3041
0.405	2402	0.560	3060
0.410	2426	0.565 2/	3079
0.415	2450	0.570	3098
0.420	2473	0.575	3116
0.425	2496	0.580	3135
0.430	2519	0.585	3154
0.435	2542	---	---

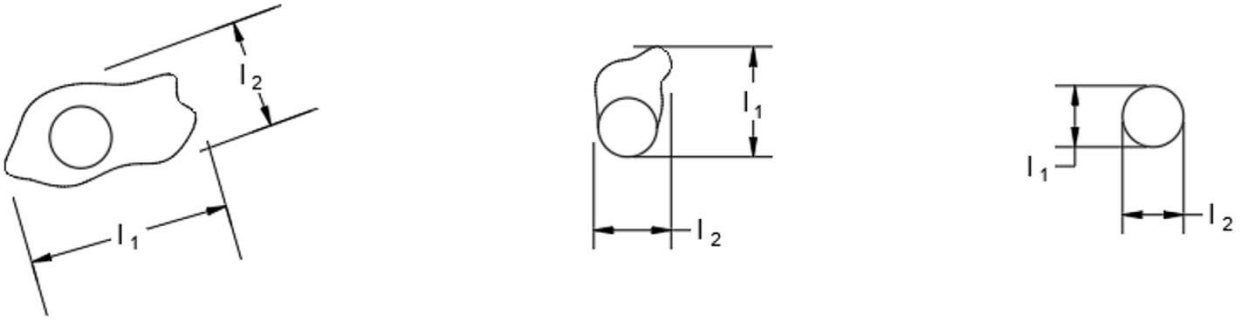
1/ Specification requirements begin for this ordered thickness

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

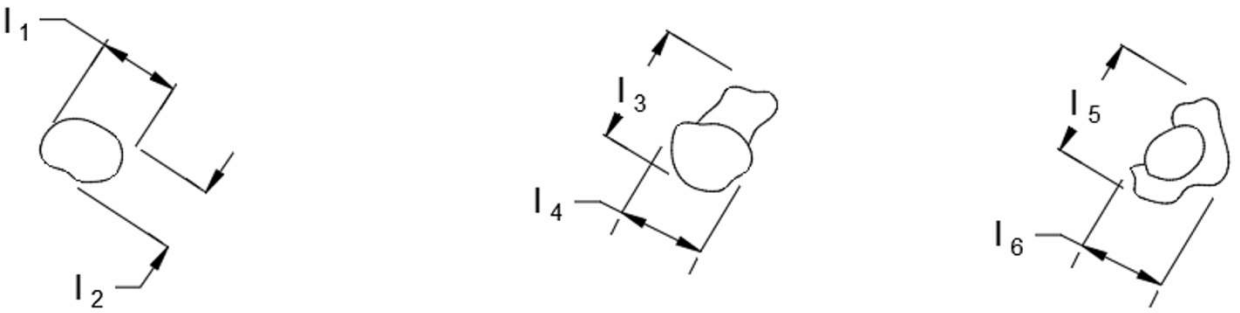
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APPENDIX A



$$Exit\ Diameter = \frac{l_1 + l_2}{2}$$

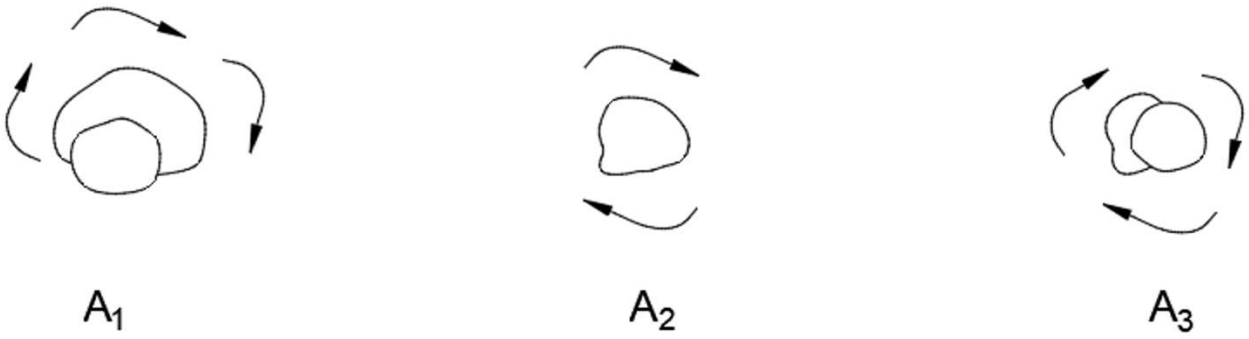
FIGURE A-1. Typical exit holes with and without back span areas.



$$\overline{D_O} = \frac{1}{3}\left[\frac{l_1 + l_2}{2} + \frac{l_3 + l_4}{2} + \frac{l_5 + l_6}{2}\right] = \frac{\sum_{i=1}^{6} l_i}{6}$$

FIGURE A-2. Illustration of computation for overall average exit diameter, $\overline{D_O}$.

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Determine each area using a planimeter around the outside perimeter of the exit hole including backspall area if any. Calculate D_O as follows:

$$\frac{\overline{A_O}}{3} = A_1 + A_2 + A_3 = \frac{\pi}{4}\overline{D_O}^2$$

$$\overline{D_O} = 2\sqrt{\frac{\overline{A_O}}{3\pi}}$$

Compare with allowable $\overline{D_O}$ allowable for the thickness (A.3.2.1 and A.3.2.2)

FIGURE A-3. Computation of overall average diameter from perimeter area measurements.

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CONCLUDING MATERIAL

Custodian:
Army - MR

Review activities:
Army - AR, AT, TE
DLA - CQ

Preparing activity:
Army - MR

(Project 9515-2023-002)

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